The Slope Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: SLOPE, FORM, WRITE and CONTEXT (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** SLOPE
- ☐ **A2** SLOPE
- ☐ **A3** FORM
- ☐ **A4** FORM
- ☐ **A5** WRITE
- ☐ **A6** CONTEXT

Entrées

10 pts each · choose 4

- ☐ **E1** SLOPE
- ☐ **E2** FORM
- ☐ **E3** WRITE
- ☐ **E4** WRITE
- ☐ **E5** CONTEXT
- ☐ **E6** SLOPE

Chef's Specials

20 pts each · choose 2

- ☐ **S1** WRITE
- ☐ **S2** CONTEXT
- ☐ **S3** FORM
- ☐ **S4** SLOPE

My skill check (circle each skill you have covered): SLOPE · FORM · WRITE · CONTEXT

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 SLOPE ☐ My pick · 5 pts

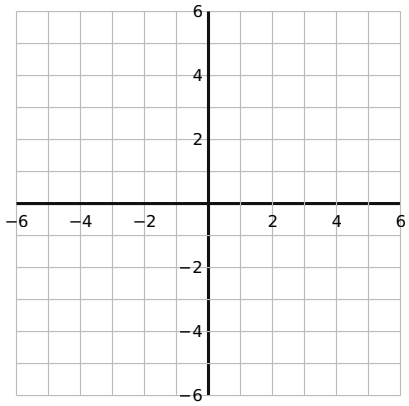
Find the slope of the line through (2, 3) and (6, 11).

A2 SLOPE ☐ My pick · 5 pts

The points (1, 9), (3, 5), and (6, −1) lie on one line. Find the slope of the line.

A3 FORM ☐ My pick · 5 pts

Graph the line $y = 2x - 1$ on the grid.



A4 FORM ☐ My pick · 5 pts

Rewrite $2x + y = 8$ in slope-intercept form and state the slope and y-intercept.

A5 WRITE ☐ My pick · 5 pts

Write the equation of the line with slope 2 that passes through (1, 5).

A6 CONTEXT ☐ My pick · 5 pts

A candle's height in cm after t hours is $h = 20 - 2t$. In a sentence each, explain what the 20 and the −2 mean.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 SLOPE ☐ My pick · 10 pts

The line through (1, k) and (5, 9) has slope $\frac{3}{2}$.

Find k and show your work.

E2 FORM ☐ My pick · 10 pts

Rewrite $3x - 4y = 20$ in slope-intercept form. State the slope, the y-intercept, and the x-intercept.

E3 WRITE ☐ My pick · 10 pts

Write the equation of the line through $(-2, 7)$ and $(4, -5)$ in slope-intercept form.

E4 WRITE ☐ My pick · 10 pts

Write the equation of the line that passes through $(-1, 5)$ and is **parallel** to $4x - 2y = 6$.

E5 CONTEXT ☐ My pick · 10 pts

A phone plan has a one-time activation fee plus a fixed monthly charge. After 2 months Ben has paid \$110 in total; after 5 months, \$215. Find the monthly charge and the activation fee, and write an equation for the total cost C after m months.

E6 SLOPE ☐ My pick · 10 pts

Tara found the slope through $(-2, 5)$ and $(4, -7)$ like this: $m = \frac{-7 - 5}{-2 - 4} = \frac{-12}{-6} = 2$.

Find her mistake, give the correct slope, and say what the sign of the correct slope tells you about the line.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 WRITE ☐ My pick · 20 pts

Line L passes through $(-4, 1)$ and $(2, 4)$.

(a) Find the slope of L.

(b) Write the equation of the line **perpendicular** to L through $(6, -3)$.

(c) Where does your line cross the x-axis?

S2 CONTEXT ☐ My pick · 20 pts

A tank drains at a steady rate. Its depth is 42 cm after 3 minutes and 27 cm after 8 minutes.

(a) Find the slope.

(b) Write an equation for the depth d after t minutes.

(c) Explain what the slope and the intercept mean.

(d) When will the tank be empty?

S3 FORM ☐ My pick · 20 pts

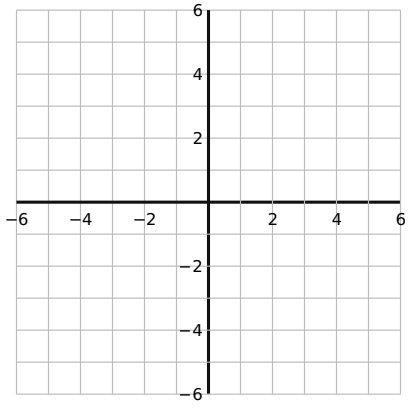
Graph $2x + 3y = 12$ on the grid.

(a) Find both intercepts.

(b) Find the slope.

(c) Draw the line.

(d) Is (3, 2) on the line? Show how you know.



S4 SLOPE ☐ My pick · 20 pts

The points A$(-2, 1)$, B$(1, k)$, and C$(4, 7)$ lie on one line.

(a) Find k using slopes.

(b) Write the equation of the line.

(c) Explain why the slope between A and B must equal the slope between B and C.